

Figure Formation & Analysis

COMPLETE LEARNING GUIDE

What You'll Learn

Figure formation and analysis questions give 3-5 separate irregular pieces and ask which target shape can be exactly formed by fitting all the given pieces together edge-to-edge, with no gaps and no overlaps — plus the reverse task, given a complete figure, identifying which set of pieces it was originally made from. This sub-module builds a systematic assembly method: matching edge lengths before anything else, using total area as a fast elimination filter, testing pieces rotated and flipped from their shown orientation, checking angle compatibility at every join, and confirming a finished assembly has zero leftover gaps or overlaps.

1. The Core Task: Assembling Given Pieces Into a Target Shape

Figure-formation questions show 3-5 separate irregular pieces and ask which target shape can be exactly formed by fitting all the given pieces together edge-to-edge, with no gaps and no overlaps — the reliable approach is to treat it like a puzzle: check compatibility systematically rather than trying to 'eyeball' whether pieces look like they'd fit a target's silhouette.

- Three pieces are given: a right triangle, a square, and a smaller right triangle. A target option shows a larger rectangle. Checking systematically: the square provides one clean edge matching the rectangle's height, and the two right triangles' hypotenuses can join to form the rectangle's remaining length exactly — confirming these 3 pieces do form that rectangle.

2. Matching Edge Lengths Before Anything Else

Before considering how pieces might visually arrange, measure each piece's edges against the other pieces' edges and against the target shape's edges — two pieces can only be joined along edges of EQUAL length, so any piece with no edge matching any other piece's edge length cannot be validly joined to it at all.

- Piece A has edges of relative lengths 3, 4, 5 (a right triangle). Piece B has edges of relative lengths 3, 3, 3, 3 (a square). Piece A's '3' edge and Piece B's '3' edge are compatible for joining; Piece A's '4' or '5' edges have no match on Piece B and cannot be the shared join edge between these two specific pieces.

3. Checking Total Area Consistency

As a fast filter before detailed assembly, roughly estimate whether the combined area of all the given pieces plausibly equals the target shape's area — if the pieces are visibly too small (or too large) in total to cover the target shape, that target option can be eliminated immediately without attempting a detailed fit.

- Three small pieces, each roughly the size of a small triangle, are given. One target option is a large hexagon clearly several times bigger than the combined size of the 3 pieces. This option is eliminated on area grounds alone — no valid arrangement of 3 small pieces can cover a much larger target shape.

4. Rotating and Flipping Pieces to Test Fit

Pieces are not restricted to their originally shown orientation — any piece may need to be rotated or flipped (mirrored) to fit correctly into the target shape, so when a piece doesn't seem to fit in its given orientation, systematically test it rotated by 90°, 180°, 270°, and also flipped, before concluding it doesn't belong in that arrangement.

- A trapezoid piece is shown with its longer parallel side at the bottom. In its given orientation, it doesn't fit against the other pieces to complete the target shape. Flipping it upside down allows its slanted edges to align correctly with the adjacent pieces — the piece was usable all along, just not in its originally displayed orientation.

5. Using Angle Compatibility at Joins

Beyond matching edge lengths, the ANGLES at a joined corner must also add up correctly to match the target shape's actual corner angle at that point — two pieces with matching edge lengths but incompatible angles at the join will leave a gap or an overlap at that corner even though the edge itself lines up.

- Two pieces join along a matching edge. At one end of that edge, Piece A contributes a 50° angle and Piece B contributes a 40° angle, summing to exactly 90° — matching the target shape's right-angle corner at that point, confirming a valid, gap-free join.

6. Eliminating Piece Sets With Leftover or Missing Area

After a tentative assembly, check for two failure signs: leftover overlapping material, or an uncovered gap inside the target's boundary — a valid formation has zero overlap and zero gaps, using every part of every given piece exactly once.

- A tentative assembly of 4 pieces into a target square leaves a small triangular gap uncovered in one corner. This confirms the 4 given pieces do NOT correctly form that target square — either a different target option is correct, or the pieces must be rearranged to eliminate that specific gap.

7. The Reverse Task: Breaking a Given Figure Into Its Source Pieces

The reverse question type gives one complete figure and asks which set of separate pieces it was originally cut from — apply the identical checks in reverse: the candidate piece set's total area must equal the given figure's area, and the pieces' edges must be capable of reconstructing every edge and angle of the given figure exactly.

- A given figure is an irregular pentagon. One candidate piece set (a triangle plus a quadrilateral) has a combined area matching the pentagon's area, and their edges/angles can be arranged to reconstruct the pentagon's exact outline. A second candidate set (two triangles) has the right total area but no valid edge-matching arrangement reconstructs the pentagon's specific angles — the first set is correct, not the second.

8. Worked Full Example: Assembling Four Pieces Into a Target Arrow

Combine every check in sequence for a full assembly problem: filter by total area first, filter by edge-length compatibility second, test rotations/flips third, and confirm angle compatibility and zero gaps/overlap last, before finalizing which target shape the given pieces form.

- Four pieces (two right triangles and two rectangles) are given, with a target option shaped like a thick arrow. Area check passes. Edge check passes: each piece has at least one edge matching an adjacent piece's edge. Rotation test: one rectangle must be rotated 90° to align with the arrow's shaft. Angle check: all joined corners sum correctly to the arrow's actual corner angles. No gaps or overlaps remain — confirming these 4 pieces correctly form the target arrow shape.

Common Pitfalls

- Judging whether pieces fit a target shape by overall visual impression instead of systematically checking edge lengths, area, and angles.
- Forgetting that pieces may need to be rotated or flipped (mirrored) from their originally shown orientation to fit correctly.
- Matching edge LENGTHS between two pieces while ignoring whether the ANGLES at that join actually sum correctly for a gap-free fit.
- Not eliminating an obviously wrong target option early using a fast total-area comparison before attempting a detailed assembly.
- Accepting a tentative assembly that has a small leftover gap or overlap, rather than treating any gap/overlap as an automatic disqualification.
- In the reverse (breakdown) task, matching only the total area of a candidate piece set without confirming its edges can actually reconstruct the given figure's exact angles.
- Assuming every given piece must be used in a simple perimeter order rather than testing multiple possible join combinations.

Key Takeaways

- Check compatibility systematically: total area first (fast elimination), then edge-length matching, then angle compatibility at each join.
- Any piece may need rotation or flipping (mirroring) from its shown orientation — don't reject a piece as non-fitting too early.
- Matching edge lengths alone isn't enough; the angles at each join must also sum correctly to match the target's actual corners.
- A valid formation has zero leftover gaps and zero overlaps, using every part of every piece exactly once.
- For the reverse (breakdown) task, apply the identical area/edge/angle checks in reverse to confirm a candidate piece set.
- Use a fast area comparison to eliminate obviously mismatched target options before attempting a full detailed assembly.

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